



```

name: SimulationBaltic
log: C:\Users\hansg\Documents\bestanden\PhD ebs\JRF work Stata\SHARE data 7\ou
> tput.smcl
log type: smcl
opened on: 25 Aug 2026, 11:19:12

1 .
2 . asdoc, text(\par \qc Financial Sustainability of the Care for the Disabled: A Simula
> tion of Benefits for Older People in the Baltic States up to 2036 with data from SHA
> RE and Eurostat) fs(12) save(report.doc) replace
(file report.doc not found)
Click to Open File: report.doc

3 . asdoc, text(\par \qc Hans Gevers - Junior Research Fellow at the Estonian Business S
> chool https://orcid.org/0009-0001-0249-4142 hans.gevers@ebs.ee) fs(10) save(report.
> doc) append
Click to Open File: report.doc

4 .
5 . *>>>PREPROCESSING
6 .
7 . foreach num of numlist 8(1)9{
2. clear all
3. cd "C:\users\hansg\Documents\bestanden\PhD ebs\JRF work Stata\SHARE data
> 7\sharew`num'_rel9-0-0_ALL_datasets_stata\"
4. use "sharew`num'_rel9-0-0_gv_imputations.dta"
5. keep if implicat==1
6. merge 1:1 mergeid using "sharew`num'_rel9-0-0_gv_health.dta"
7. isid mergeid
8. drop _merge
9. merge 1:1 mergeid using "sharew`num'_rel9-0-0_gv_weights.dta"
10. isid mergeid
11. drop _merge
12.
8 . if `num'==8{
13. gen Qyear=2020
14. }
15. if `num'==9{
16. gen Qyear=2022
17. }
18.
9 . keep if country==57 | country==48 | country==35
19.
10. save "C:\users\hansg\Documents\bestanden\PhD ebs\JRF work Stata\SHARE data 7
> \W`num'_final.dta", replace
20. }
C:\users\hansg\Documents\bestanden\PhD ebs\JRF work Stata\SHARE data 7\sharew8_rel9-0-
> 0 ALL_datasets_stata
(214,780 observations deleted)
(label orienti already defined)
(label eurod already defined)
(label numeracy already defined)
(label gali already defined)
(label language already defined)
(label country already defined)

Result Number of obs
Not matched 0
Matched 53,695 (_merge==3)

(label country already defined)
(label language already defined)

```

Result	Number of obs
Not matched	0
Matched	53,695 (_merge==3)

(47,632 observations deleted)

file C:\users\hansg\Documents\bestanden\PhD ebs\JRF work Stata\SHARE data
7\W8_final.dta saved

C:\users\hansg\Documents\bestanden\PhD ebs\JRF work Stata\SHARE data 7\sharew9_rel9-0-

> 0 ALL datasets stata

(277,788 observations deleted)

(label **orienti** already defined)

(label **eurod** already defined)

(label **numeracy** already defined)

(label **gali** already defined)

(label **language** already defined)

(label **country** already defined)

Result	Number of obs
Not matched	0
Matched	69,447 (_merge==3)

(label **country** already defined)

(label **language** already defined)

Result	Number of obs
Not matched	0
Matched	69,447 (_merge==3)

(61,831 observations deleted)

file C:\users\hansg\Documents\bestanden\PhD ebs\JRF work Stata\SHARE data
7\W9_final.dta saved

11.

12. clear all

13. cd "C:\users\hansg\Documents\bestanden\PhD ebs\JRF work Stata\SHARE data 7\
C:\users\hansg\Documents\bestanden\PhD ebs\JRF work Stata\SHARE data 7"

14. use "W8_final.dta"

15. merge 1:1 mergeid using "weightslong.dta", keep(match)
(label **country** already defined)

Result	Number of obs
Not matched	0
Matched	5,101 (_merge==3)

16. drop _merge

17. save "W8_final.dta", replace
file **W8_final.dta** saved

18.

19. clear all

```
20. cd "C:\users\hansg\Documents\bestanden\PhD ebs\JRF work Stata\SHARE data 7\"
    C:\users\hansg\Documents\bestanden\PhD ebs\JRF work Stata\SHARE data 7
```

```
21. use "W9_final.dta"
```

```
22. merge 1:1 mergeid using "weightslong.dta", keep(match)
    (label country already defined)
```

Result	Number of obs	
Not matched	0	
Matched	4,921	(_merge ==3)

```
23. drop _merge
```

```
24. save "W9_final.dta", replace
    file W9_final.dta saved
```

```
25.
```

```
26. clear all
```

```
27. cd "C:\users\hansg\Documents\bestanden\PhD ebs\JRF work Stata\SHARE data 7\"
    C:\users\hansg\Documents\bestanden\PhD ebs\JRF work Stata\SHARE data 7
```

```
28. use "W8_final.dta"
```

```
29. append using "W9_final.dta"
    (variable psu was str10, now str11 to accommodate using data's values)
    (variable ssu was str6, now str13 to accommodate using data's values)
    (label pstat already defined)
    (label language already defined)
    (label country already defined)
    (label Hrespt already defined)
    (label pcskill already defined)
    (label IMPflag already defined)
    (label orienti already defined)
    (label bfi10 already defined)
    (label thospital already defined)
    (label satisfied already defined)
    (label often already defined)
    (label yedu already defined)
    (label fdistress already defined)
    (label writing already defined)
    (label reading already defined)
    (label hearing already defined)
    (label eyesight already defined)
    (label mstat already defined)
    (label sphus already defined)
    (label noyes already defined)
    (label hnrsc already defined)
    (label undersq already defined)
    (label clarif already defined)
    (label willans already defined)
    (label otrf already defined)
    (label empstat already defined)
    (label cjs already defined)
    (label isced already defined)
    (label gender already defined)
    (label num already defined)
    (label gali already defined)
    (label meat already defined)
    (label legeggs already defined)
    (label fruit already defined)
    (label numeracy already defined)
    (label numeracy2 already defined)
    (label eurod already defined)
    (label dairyp already defined)
    (label weekday already defined)
    (label specday already defined)
    (label dkrf already defined)
    (label dummi already defined)
```

```

(label zero_one already defined)
(label loneli already defined)
(label chronic2 already defined)
(label sphus2 already defined)
(label bmi2 already defined)
(label mobilite2 already defined)
(label mobilite3 already defined)
(label adl2 already defined)
(label iadl2 already defined)
(label phactiv already defined)

30.
31. save "dataset.dta", replace
    file dataset.dta saved

32.
33. clear all

34.
35. use "dataset.dta"

36.
37. *>>>PREPARATION
38.
39. label define isc 0 "None" 1 "Primary education" 2 "Lower secondary education" 3 "Upper
    > secondary education" 4 "Post-secondary non-tertiary education" 5 "Bachelor's or e
    > quivalent level" 6 "Master's or equivalent level/Doctoral or equivalent level"

40. label values isced isc

41. label variable isced "Education"

42.
43. generate income = thinc-ypen3

44.
45. egen id= group(mergeid)

46.
47. drop if cjs== -99
    (70 observations deleted)

48.
49. generate educ=0

50. replace educ=1 if isced==3 | isced==4
    (5,234 real changes made)

51. replace educ=2 if isced>=5
    (2,943 real changes made)

52. label define educL 0 "Less than primary, primary and lower secondary education (ISCE
    > D levels 0-2)" 1 "Upper secondary and post-secondary non-tertiary education (ISCED l
    > evels 3 and 4)" 2 "Tertiary education (ISCED levels 5-8)"

53. label values educ educL

54.
55. keep single ypen3 gender cjs age educ sphus country income gali id Qyear sphus

```

56.
 57. drop if ypen3>7000
 (7 observations deleted)

58.
 59. *>>>DESCRIPTIVES
 60.
 61. tabulate country

Country identifier	Freq.	Percent	Cum.
Estonia	5,520	55.51	55.51
Lithuania	2,422	24.35	79.86
Latvia	2,003	20.14	100.00
Total	9,945	100.00	

62.
 63. asdoc codebook single ypen3 gender cjs age educ sphus country income gali id Qyear,
 > compact save(report.doc) append

Variable	Obs	Unique	Mean	Min	Max	Label
single	9945	2	.4113625	0	1	Single
ypen3	9945	378	232.8604	0	6816	Disability pensions/benefits
gender	9945	2	1.631875	1	2	Gender
cjs	9945	6	1.949321	1	97	Current job situation
age	9945	64	69.68678	34	100	Age
educ	9945	3	1.117748	0	2	
sphus	9945	5	3.740875	1	5	Self-perceived health - US scale
country	9945	3	42.59698	35	57	Country identifier
income	9945	4450	10096.15	-.0000737	170698	
gali	9945	2	.5882353	0	1	Limitation with activities
id	9945	5086	2559.401	1	5101	group (mergeid)
Qyear	9945	2	2020.979	2020	2022	

Click to Open File: [report.doc](#)

64.
 65. asdoc codebook, save(report.doc) append

country	Country identifier		
Type: Numeric (byte)			
Label: country			
Range: [35,57]		Units: 1	
Unique values: 3		Missing .: 0/9,945	
Tabulation: Freq. Numeric Label			
5,520 35 Estonia			
2,422 48 Lithuania			
2,003 57 Latvia			
single	Single		
Type: Numeric (byte)			
Label: noyes			
Range: [0,1]		Units: 1	
Unique values: 2		Missing .: 0/9,945	

Tabulation:	Freq.	Numeric	Label
	5,854	0	No
	4,091	1	Yes

ypen3**Disability pensions/benefits**

Type: Numeric (**double**)

Range: [0,6816] Units: 1.000e-07
Unique values: 378 Missing .: 0/9,945

Mean: 232.86
Std. dev.: 884.674

Percentiles:	10%	25%	50%	75%	90%
	0	0	0	0	276

gender**Gender**

Type: Numeric (**byte**)
Label: **gender**

Range: [1,2] Units: 1
Unique values: 2 Missing .: 0/9,945

Tabulation:	Freq.	Numeric	Label
	3,661	1	Male
	6,284	2	Female

age**Age**

Type: Numeric (**int**)
Label: **num**, but 64 nonmissing values are not labeled

Range: [34,100] Units: 1
Unique values: 64 Missing .: 0/9,945

Examples: 60
66
72
80

sphus**Self-perceived health - US scale**

Type: Numeric (**byte**)
Label: **sphus**

Range: [1,5] Units: 1
Unique values: 5 Missing .: 0/9,945

Tabulation:	Freq.	Numeric	Label
	154	1	Excellent
	416	2	Very good
	2,842	3	Good
	4,974	4	Fair
	1,559	5	Poor

gali**Limitation with activities**

Type: Numeric (**byte**)
Label: **gali**

Range: [0,1] Units: 1
 Unique values: 2 Missing .: 0/9,945

Tabulation:	Freq.	Numeric	Label
	4,095	0	Not limited
	5,850	1	Limited

cjs Current job situation

Type: Numeric (**byte**)
 Label: **cjs**

Range: [1,97] Units: 1
 Unique values: 6 Missing .: 0/9,945

Tabulation:	Freq.	Numeric	Label
	6,180	1	Retired
	2,947	2	Employed or self-employed
	241	3	Unemployed
	436	4	Permanently sick
	96	5	Homemaker
	45	97	Other

Qyear (unlabeled)

Type: Numeric (**float**)

Range: [2020,2022] Units: 1
 Unique values: 2 Missing .: 0/9,945

Tabulation:	Freq.	Value
	5,075	2020
	4,870	2022

income (unlabeled)

Type: Numeric (**float**)

Range: [-.00007374,170697.95] Units: 1.000e-20
 Unique values: 4,450 Missing .: 0/9,945

Mean: 10096.1
 Std. dev.: 8970.94

Percentiles:	10%	25%	50%	75%	90%
	2000	4920	7800	12898.5	20000

id group (mergeid)

Type: Numeric (**float**)

Range: [1,5101] Units: 1
 Unique values: 5,086 Missing .: 0/9,945

Mean: 2559.4
 Std. dev.: 1474.54

Percentiles:	10%	25%	50%	75%	90%
	512	1281	2565	3838	4597

educ (unlabeled)

Type: Numeric (**float**)
Label: **educL**

Range: [0,2]
Unique values: 3

Units: 1
Missing .: 0/9,945

Tabulation:	Freq.	Numeric	Label
	1,772	0	Less than primary, primary and lower secondary education (ISCED levels 0-2)
	5,230	1	Upper secondary and post-secondary non-tertiary education (ISCED levels 3 and 4)
	2,943	2	Tertiary education (ISCED levels 5-8)

Click to Open File: [report.doc](#)

- 66.
67. hist ypen3 if ypen3>0, scheme(s2color) xlabel(0(1000)7000,angle(0) labsize(small) gr
> id) ylabel(0.000(0.0002)0.001,angle(0) labsize(small) grid format(%9.0gc))
(bin=30, start=40, width=225.86667)
68. graph export benefits.png, replace height(2400)
(file **benefits.png** not found)
file **benefits.png** saved as PNG format
69. vioplot age, over(gender) over(country) scheme(s2color) ytitle("Number of respondent
> s", margin(small)) ylabel(30(10)100,angle(0) labsize(small) grid)
70. graph export sample.png, replace height(2400)
(file **sample.png** not found)
file **sample.png** saved as PNG format
71. vioplot age if ypen3>0, over(gender) over(country) scheme(s2color) ytitle("Number of
> respondents", margin(small)) ylabel(30(10)100,angle(0) labsize(small) grid)
72. graph export sampleDis.png, replace height(2400)
(file **sampleDis.png** not found)
file **sampleDis.png** saved as PNG format
- 73.
74. asdoc spearman single ypen3 gender cjs age educ sphus country income gali id Qyear s
> phus, save(report.doc) append

Number of observations = 9,945

>	income	single gali	ypen3 id	gender Qyear	cjs sphus	age	educ	sphus	country
	single	1.0000							
	ypen3	-0.0097	1.0000						
	gender	0.3012	-0.0174	1.0000					
	cjs	-0.1691	0.2510	-0.0667	1.0000				
	age	0.2609	-0.1638	0.0717	-0.7428	1.0000			
	educ	-0.1050	-0.0654	0.0452	0.1552	-0.2079	1.0000		
	sphus	0.0936	0.2095	-0.0052	-0.2444	0.3377	-0.2321	1.0000	
	country	-0.0444	-0.0934	-0.0116	0.0281	-0.0854	0.0262	-0.0231	1.0000
>	income	-0.4996	-0.1248	-0.1491	0.0401	-0.0771	0.2030	-0.1564	-0.3446
>	1.0000								
>	gali	0.0862	0.2267	0.0206	-0.2104	0.2914	-0.1698	0.5395	-0.0195
>	-0.1439	1.0000							
>	id	-0.0447	-0.0920	-0.0068	0.0232	-0.0732	0.0286	-0.0242	0.8980
>	-0.3005	-0.0242	1.0000						
>	Qyear	0.0285	-0.0355	0.0137	-0.0633	0.0854	0.0050	0.0145	0.0048
>	0.0791	0.0255	0.0041	1.0000					
>	sphus	0.0936	0.2095	-0.0052	-0.2444	0.3377	-0.2321	1.0000	-0.0231
>	-0.1564	0.5395	-0.0242	0.0145	1.0000				

Click to Open File: [report.doc](#)


```

75.
76. *>>>ANALYSIS
77.
78. xtset id Qyear

Panel variable: id (unbalanced)
Time variable: Qyear, 2020 to 2022, but with gaps
Delta: 1 unit

79.
80. *-----
81. *ESTONIA
82.
83. *general model
84.
85. xttobit ypen3 i.gender age i.gali i.educ income if country==35, ll(0) iterate(25) ba
> selevels tobit

```

Fitting comparison model:

Fitting constant-only model:

```

Iteration 0: Log likelihood = -12985.23
Iteration 1: Log likelihood = -8884.6955
Iteration 2: Log likelihood = -8760.2216
Iteration 3: Log likelihood = -8757.2061
Iteration 4: Log likelihood = -8757.2031
Iteration 5: Log likelihood = -8757.2031

```

Fitting full model:

```

Iteration 0: Log likelihood = -12670.992
Iteration 1: Log likelihood = -8879.4249
Iteration 2: Log likelihood = -8317.5857
Iteration 3: Log likelihood = -8259.0713
Iteration 4: Log likelihood = -8258.2286
Iteration 5: Log likelihood = -8258.2283

```

Obtaining starting values for full model:

```

Iteration 0: Log likelihood = -44867.876
Iteration 1: Log likelihood = -44820.602
Iteration 2: Log likelihood = -44819.948
Iteration 3: Log likelihood = -44819.948

```

Fitting full model:

```

Iteration 0: Log likelihood = -11829.704
Iteration 1: Log likelihood = -9335.9995 (not concave)
Iteration 2: Log likelihood = -8353.5036
Iteration 3: Log likelihood = -8115.3181
Iteration 4: Log likelihood = -8098.1776
Iteration 5: Log likelihood = -8098.1554
Iteration 6: Log likelihood = -8098.1554
Iteration 7: Log likelihood = -8098.1553

```

Random-effects tobit regression

```

Limits: Lower = 0
Upper = +inf

```

```

Group variable: id
Random effects u_i ~ Gaussian

```

Integration method: **mvaghermite**

Log likelihood = **-8098.1553**

```

Number of obs      = 5,520
Uncensored         = 761
Left-censored      = 4,759
Right-censored     = 0

```

```

Number of groups   = 2,837
Obs per group:
    min = 1
    avg = 1.9
    max = 2

```

Integration pts. = **12**

```

Wald chi2(6)       = 428.06
Prob > chi2        = 0.0000

```

```

> efficient
> Std. err.
> z
> P>|z|
> [95% con
> f. interval]

```

```

> 0
> (base)
> 71.14944
> 167.157
> 0.43
> 0.670
> -256.4722
> 398.7711
>
> 136.3039
> 8.466988
> -16.10
> 0.000
> -152.8989
> -119.709
>
> 0
> (base)
> 2984.743
> 203.0102
> 14.70
> 0.000
> 2586.85
> 3382.635
>
> Less than primary, primary and lower secondary education (ISCED levels 0-2)
> 0
> (base)
> Upper secondary and post-secondary non-tertiary education (ISCED levels 3 and 4)
> 580.6917
> 209.8049
> -2.77
> 0.006
> -991.9017
> -169.4817
> Tertiary education (ISCED levels 5-8)
> 1240.876
> 255.6194
> -4.85
> 0.000
> -1741.881
> -739.8708
>
> -.072545
> .0092093
> -7.88
> 0.000
> -.0905949
> -.0544951
>
> 5269.962
> 637.9724

```

LR test of sigma_u=0: chibar2(01) = 320.15 Prob >= chibar2 = 0.000

```
87. xttoibit ypen3 i.gender age i.gali i.educ income if country==35, ll(0) iterate(25) ba
> selevels tobit
```

Fitting constant-only model:

Fitting full model:

Obtaining starting values for full model:

Fitting full model:


```

Upper secondary and post-secondary non-tertiary education (ISCED levels 3 and 4) | -
> 580.6917
>      209.8049
>      -2.77
>      0.006
>      -991.9017
>      -169.4817
>      Tertiary education (ISCED levels 5-8) | -
> 1240.876
>      255.6194
>      -4.85
>      0.000
>      -1741.881
>      -739.8708
>
>      income |
> -.072545
>      .0092093
>      -7.88
>      0.000
>      -.0905949
>      -.0544951
>      _cons |
> 5269.962
>      637.9724
>      8.26
>      0.000
>      4019.559
>      6520.365
>
>      /sigma_u |
> 2450.199
>      108.1976
>      22.65
>      0.000
>      2238.136
>      2662.262
>      /sigma_e |
> 1844.299
>      69.26943
>      26.63
>      0.000
>      1708.534
>      1980.065
>
>      rho |
> .6383338
>      .0262833
>      .5857132
>      .6884586
>
LR test of sigma_u=0: chibar2(01) = 320.15 Prob >= chibar2 = 0.000

```

```

88. outreg2 using results.xls, excel dec(3) alpha(0.01, 0.05, 0.10) symbol(**, **, *) c
> title(CoreEE) replace
results.xls
dir : seeout

```

```

89. outreg2 using resultsNoStar.xls, excel dec(3) ctitle(CoreEE) noaster replace
   resultsNoStar.xls
   dir : seeout
90.
91. *for disability
92.
93. generate ageD=0

94. replace ageD=1 if age>=55 & age<=64
   (2,789 real changes made)

95. replace ageD=2 if age>=65 & age<=74
   (3,129 real changes made)

96. replace ageD=3 if age>=75 & age<=84
   (2,530 real changes made)

97. replace ageD=4 if age>=85
   (848 real changes made)

98. label define ageDl 0 "Less than 55 years old" 1 "Between 55 and 64 years old" 2 "Bet
   > ween 65 and 74 years old" 3 "Between 75 and 84 years old" 4 "Older than 85 years"

99. label values ageD ageDl

100
101 xttobit ypen3 i.gender i.ageD i.gali i.educ income if country==35, ll(0) iterate(25)
   > baselevels

```

Fitting comparison model:

Fitting constant-only model:

```

Iteration 0: Log likelihood = -12985.23
Iteration 1: Log likelihood = -8884.6955
Iteration 2: Log likelihood = -8760.2216
Iteration 3: Log likelihood = -8757.2061
Iteration 4: Log likelihood = -8757.2031
Iteration 5: Log likelihood = -8757.2031

```

Fitting full model:

```

Iteration 0: Log likelihood = -12567.169
Iteration 1: Log likelihood = -8672.9415
Iteration 2: Log likelihood = -8157.0625
Iteration 3: Log likelihood = -8121.2781
Iteration 4: Log likelihood = -8121.1218
Iteration 5: Log likelihood = -8121.1218

```

Obtaining starting values for full model:

```

Iteration 0: Log likelihood = -44682.34
Iteration 1: Log likelihood = -44635.023
Iteration 2: Log likelihood = -44634.354
Iteration 3: Log likelihood = -44634.354

```

Fitting full model:

```

Iteration 0: Log likelihood = -11734.867
Iteration 1: Log likelihood = -9278.6293 (not concave)
Iteration 2: Log likelihood = -8341.6104
Iteration 3: Log likelihood = -8037.9167
Iteration 4: Log likelihood = -7985.8066
Iteration 5: Log likelihood = -7984.3365
Iteration 6: Log likelihood = -7984.3305
Iteration 7: Log likelihood = -7984.3305

```

Random-effects tobit regression

Limits: Lower = 0
Upper = +infGroup variable: id
Random effects u_i ~ GaussianNumber of obs = 5,520
Uncensored = 761
Left-censored = 4,759
Right-censored = 0Number of groups = 2,837
Obs per group: min = 1
avg = 1.9
max = 2

Integration method: mvaghermite

Integration pts. = 12

Log likelihood = -7984.3305

Wald chi2(9) = 608.68
Prob > chi2 = 0.0000

						ypen3	Co
> efficient	Std. err.	z	P> z	[95% con	f. interval]		
						gender	
						Male	
> 0	(base)						
> 81.92267	152.6722	0.54	0.592	-217.3094	381.1547	Female	
						aged	
						Less than 55 years old	
> 0	(base)						
						Between 55 and 64 years old	
> 632.0103	238.2304	2.65	0.008	165.0874	1098.933		
						Between 65 and 74 years old	-
> 2835.626	265.4856	-10.68	0.000	-3355.968	-2315.284		
						Between 75 and 84 years old	-
> 3133.635	274.2828	-11.42	0.000	-3671.219	-2596.05		
						Older than 85 years	-
> 3071.775	326.5054	-9.41	0.000	-3711.714	-2431.836		

[illegible]

LR test of sigma u=0: chibar2(01) = 273.58

Prob >= chibar2 = 0.000


```

102 outreg2 using results.xls, excel dec(3) alpha(0.01, 0.05, 0.10) symbol(***, **, *) c
> title(DisabilityEE) append
   results.xls
   dir : seeout

```

```

103 outreg2 using resultsNoStar.xls, excel dec(3) ctitle(DisabilityEE) noaster append
   resultsNoStar.xls
   dir : seeout

```

```

104
105 *for education
106
107 generate ageE=0

```

```

108 replace ageE=1 if age>=55 & age<=64
    (2,789 real changes made)

```

```

109 replace ageE=2 if age>=65
    (6,507 real changes made)

```

```

110 label define agel 0 "Less than 55 years old" 1 "Between 55 and 64 years old" 2 "Olde
> r than 64 years"

```

```

111 label values ageE agel

```

```

112
113 xttobit ypen3 i.gender i.ageE i.gali i.educ income if country==35, ll(0) iterate(25)
> baselevels tobit

```

Fitting comparison model:

Fitting constant-only model:

```

Iteration 0:  Log likelihood = -12985.23
Iteration 1:  Log likelihood = -8884.6955
Iteration 2:  Log likelihood = -8760.2216
Iteration 3:  Log likelihood = -8757.2061
Iteration 4:  Log likelihood = -8757.2031
Iteration 5:  Log likelihood = -8757.2031

```

Fitting full model:

```

Iteration 0:  Log likelihood = -12573.826
Iteration 1:  Log likelihood = -8679.1524
Iteration 2:  Log likelihood = -8159.2818
Iteration 3:  Log likelihood = -8122.9056
Iteration 4:  Log likelihood = -8122.7431
Iteration 5:  Log likelihood = -8122.7431

```

Obtaining starting values for full model:

```

Iteration 0:  Log likelihood = -44685.139
Iteration 1:  Log likelihood = -44640.622
Iteration 2:  Log likelihood = -44640.061
Iteration 3:  Log likelihood = -44640.06

```

Fitting full model:

```

Iteration 0:  Log likelihood = -11733.307
Iteration 1:  Log likelihood = -9262.1334 (not concave)
Iteration 2:  Log likelihood = -8275.6704
Iteration 3:  Log likelihood = -8011.1638
Iteration 4:  Log likelihood = -7985.689
Iteration 5:  Log likelihood = -7985.5156
Iteration 6:  Log likelihood = -7985.5154

```

Random-effects tobit regression

```

Number of obs      = 5,520
Uncensored         = 761
Left-censored      = 4,759
Right-censored     = 0

```

```

Limits: Lower = 0
        Upper = +inf

```

Group variable: **id**
 Random effects $u_i \sim \text{Gaussian}$

Number of groups = **2,837**
 Obs per group:
 min = **1**
 avg = **1.9**
 max = **2**

Integration method: **mvaghermite**

Integration pts. = **12**

Log likelihood = **-7985.5154**

Wald $\chi^2(7)$ = **603.29**
 Prob > χ^2 = **0.0000**

						ypen3	Co
	efficient	Std. err.	z	P> z	[95% con f. interval]		
						gender	
						Male	
>	0						
>	(base)						
>	69.05354	152.4479	0.45	0.651	-229.7388	367.8459	
>							
>							
>							
>							
						ageE	
						Less than 55 years old	
>	0						
>	(base)						
						Between 55 and 64 years old	
>	626.5454	238.3319	2.63	0.009	159.4235	1093.667	
>							
>							
>							
>							
>							
>	2979.174	249.9052	-11.92	0.000	-3468.979	-2489.369	
>							
>							
>							
>							
						gali	
						Not limited	
>	0						
>	(base)						
						Limited	
>	2958.171	191.4826	15.45	0.000	2582.872	3333.47	
>							
>							
>							
>							
						educ	
						Less than primary, primary and lower secondary education (ISCED levels 0-2)	
>	0						
>	(base)						
						Upper secondary and post-secondary non-tertiary education (ISCED levels 3 and 4)	
>	475.8512	187.452					
>							

>			-2.54	0.011	-843.2504	-108.452			
>									
>									
>	1297.553								
>		233.3868							
>			-5.56						
>				0.000					
>					-1754.983	-840.1231			
>									
>	.0587093							income	-
>		.0082588							
>			-7.11						
>				0.000					
>					-.0748963	-.0425223			
>									
>	2351.857							_cons	-
>		346.1047							
>			-6.80						
>				0.000					
>					-3030.21	-1673.505			
>									
>								/sigma_u	
>	2178.042								
>		98.30976							
>			22.15						
>				0.000					
>					1985.359				
>						2370.726			
>								/sigma_e	
>	1708.686								
>		64.92615							
>			26.32						
>				0.000					
>					1581.433				
>						1835.939			
>									
>									
>	.6190231							rho	
>		.0280908							
>					.5629409				
>						.6727077			
>									

LR test of sigma_u=0: chibar2(01) = 274.46 Prob >= chibar2 = 0.000

```
114 outreg2 using results.xls, excel dec(3) alpha(0.01, 0.05, 0.10) symbol(**, **, *) c
> title(EducationEE) append
results.xls
dir : seeout
```

```
115 outreg2 using resultsNoStar.xls, excel dec(3) ctitle(EducationEE) noaster append
    resultsNoStar.xls
    dir : seeout
```

```

116
117 *for income
118
119 xttobit ypen3 i.gender i.ageE i.gali i.educ income if country==35, ll(0) iterate(25)
> baselevels tobit

```

Fitting comparison model:

Fitting constant-only model:

```

Iteration 0: Log likelihood = -12985.23
Iteration 1: Log likelihood = -8884.6955
Iteration 2: Log likelihood = -8760.2216
Iteration 3: Log likelihood = -8757.2061
Iteration 4: Log likelihood = -8757.2031
Iteration 5: Log likelihood = -8757.2031

```

Fitting full model:

```

Iteration 0: Log likelihood = -12573.826
Iteration 1: Log likelihood = -8679.1524
Iteration 2: Log likelihood = -8159.2818
Iteration 3: Log likelihood = -8122.9056
Iteration 4: Log likelihood = -8122.7431
Iteration 5: Log likelihood = -8122.7431

```

Obtaining starting values for full model:

```

Iteration 0: Log likelihood = -44685.139
Iteration 1: Log likelihood = -44640.622
Iteration 2: Log likelihood = -44640.061
Iteration 3: Log likelihood = -44640.06

```

Fitting full model:

```

Iteration 0: Log likelihood = -11733.307
Iteration 1: Log likelihood = -9262.1334 (not concave)
Iteration 2: Log likelihood = -8275.6704
Iteration 3: Log likelihood = -8011.1638
Iteration 4: Log likelihood = -7985.689
Iteration 5: Log likelihood = -7985.5156
Iteration 6: Log likelihood = -7985.5154

```

Random-effects tobit regression

```

Limits: Lower = 0
Upper = +inf

```

```

Group variable: id
Random effects u_i ~ Gaussian

```

Integration method: mvaghermite

Log likelihood = -7985.5154

```

Number of obs      = 5,520
Uncensored         = 761
Left-censored      = 4,759
Right-censored     = 0

```

```

Number of groups   = 2,837
Obs per group:
    min = 1
    avg = 1.9
    max = 2

```

Integration pts. = 12

```

Wald chi2(7)      = 603.29
Prob > chi2       = 0.0000

```

```
> efficient
```

```
> Std. err.
```

```
> z
```

```
> P>|z|
```

```
> [95% con
```

```
> f. interval]
```

ypen3 | Co

gender
Male

```

>      0
>      (base)
>      Female |
> 69.05354
>      152.4479
>      0.45
>      0.651
>      -229.7388
>      367.8459
>
>      ageE
>      Less than 55 years old |
>      0
>      (base)
>      Between 55 and 64 years old |
> 626.5454
>      238.3319
>      2.63
>      0.009
>      159.4235
>      1093.667
>      Older than 64 years | -
> 2979.174
>      249.9052
>      -11.92
>      0.000
>      -3468.979
>      -2489.369
>
>      gali
>      Not limited |
>      0
>      (base)
>      Limited |
> 2958.171
>      191.4826
>      15.45
>      0.000
>      2582.872
>      3333.47
>
>      educ
>      Less than primary, primary and lower secondary education (ISCED levels 0-2) |
>      0
>      (base)
>      Upper secondary and post-secondary non-tertiary education (ISCED levels 3 and 4) | -
> 475.8512
>      187.452
>      -2.54
>      0.011
>      -843.2504
>      -108.452
>      Tertiary education (ISCED levels 5-8) | -
> 1297.553
>      233.3868
>      -5.56
>      0.000
>      -1754.983
>      -840.1231
>
>      income | -
> .0587093
>      .0082588
>      -7.11
>      0.000
>      -.0748963
>      -.0425223
>
>      _cons | -
> 2351.857
>      346.1047
>      -6.80
>      0.000

```

```
120 outreg2 using results.xls, excel dec(3) alpha(0.01, 0.05, 0.10) symbol(**, **, *) c
> title(IncomeEE) append
results.xls
dir : seeout
```

```
121 outreg2 using resultsNoStar.xls, excel dec(3) ctitle(IncomeEE) noaster append
    resultsNoStar.xls
    dir : seeout
```

```

122
123 *-----
124 *LITHUANIA
125
126 *general model
127
128 xttobit ypen3 i.gender age i.gali i.educ income if country==48, ll(0) iterate(25) ba
    > selevels tobit

```

Fitting comparison model:

Fitting constant-only model:

```
Iteration 0: Log likelihood = -4852.3056
Iteration 1: Log likelihood = -2746.6214
Iteration 2: Log likelihood = -2638.7788
Iteration 3: Log likelihood = -2635.3474
Iteration 4: Log likelihood = -2635.3462
Iteration 5: Log likelihood = -2635.3462
```

Fitting full model:

```
Iteration 0: Log likelihood = -4782.9393
Iteration 1: Log likelihood = -2568.3998
Iteration 2: Log likelihood = -2500.1795
Iteration 3: Log likelihood = -2497.6576
Iteration 4: Log likelihood = -2497.648
Iteration 5: Log likelihood = -2497.648
```

Obtaining starting values for full model:

Iteration 0: Log likelihood = **-19455.305**
 Iteration 1: Log likelihood = **-19447.281**
 Iteration 2: Log likelihood = **-19447.254**
 Iteration 3: Log likelihood = **-19447.254**

Fitting full model:

Iteration 0: Log likelihood = **-4425.8291**
 Iteration 1: Log likelihood = **-3065.691** (not concave)
 Iteration 2: Log likelihood = **-2495.2025** (not concave)
 Iteration 3: Log likelihood = **-2464.7645**
 Iteration 4: Log likelihood = **-2432.2161**
 Iteration 5: Log likelihood = **-2429.5931**
 Iteration 6: Log likelihood = **-2429.5414**
 Iteration 7: Log likelihood = **-2429.5412**
 Iteration 8: Log likelihood = **-2429.54**
 Iteration 9: Log likelihood = **-2429.54**

Random-effects tobit regression

Limits: Lower = **0**
 Upper = **+inf**

Group variable: **id**
 Random effects $u_i \sim \text{Gaussian}$

Number of obs = **2,422**
 Uncensored = **217**
 Left-censored = **2,205**
 Right-censored = **0**

Number of groups = **1,235**
 Obs per group:
 min = **1**
 avg = **2.0**
 max = **2**

Integration method: **mvaghermite**

Integration pts. = **12**

Log likelihood = **-2429.54**

Wald chi2(6) = **101.23**
 Prob > chi2 = **0.0000**

						ypen3	Co
> efficient							
> Std. err.							
> z							
> P> z							
> [95% con							
> f. interval]							
						gender	
						Male	
> 0							
> (base)						Female	-
> 1237.641							
> 415.1885							
> -2.98							
> 0.003							
> -2051.396							
> -423.887							
						age	-
> 199.2338							
> 27.19966							
> -7.32							
> 0.000							
> -252.5441							
> -145.9234							
						gali	
> 0						Not limited	
> (base)							
						Limited	
> 3080.019							
> 422.0995							


```

129 outreg2 using results.xls, excel dec(3) alpha(0.01, 0.05, 0.10) symbol(**, **, *) c
> title(CoreLT) append
results.xls
dir : seeout

130 outreg2 using resultsNoStar.xls, excel dec(3) ctitle(CoreLT) noaster append
resultsNoStar.xls
dir : seeout

131
132 *for disability
133
134 xttobit ypen3 i.gender i.ageD i.gali i.educ income if country==48, ll(0) iterate(25)
> baselevels tobit

```

Fitting comparison model:

Fitting constant-only model:

```

Iteration 0: Log likelihood = -4852.3056
Iteration 1: Log likelihood = -2746.6214
Iteration 2: Log likelihood = -2638.7788
Iteration 3: Log likelihood = -2635.3474
Iteration 4: Log likelihood = -2635.3462
Iteration 5: Log likelihood = -2635.3462

```

Fitting full model:

```

Iteration 0: Log likelihood = -4762.4526
Iteration 1: Log likelihood = -2552.2572
Iteration 2: Log likelihood = -2471.6214
Iteration 3: Log likelihood = -2468.4818
Iteration 4: Log likelihood = -2468.4724
Iteration 5: Log likelihood = -2468.4724

```

Obtaining starting values for full model:

```

Iteration 0: Log likelihood = -19435.017
Iteration 1: Log likelihood = -19426.702
Iteration 2: Log likelihood = -19426.674

```

Fitting full model:

```

Iteration 0: Log likelihood = -4420.3106
Iteration 1: Log likelihood = -2824.5512 (not concave)
Iteration 2: Log likelihood = -2482.2688 (not concave)
Iteration 3: Log likelihood = -2458.8406 (not concave)
Iteration 4: Log likelihood = -2442.6107
Iteration 5: Log likelihood = -2413.0602
Iteration 6: Log likelihood = -2410.3349
Iteration 7: Log likelihood = -2409.7862
Iteration 8: Log likelihood = -2409.7862 (backed up)
Iteration 9: Log likelihood = -2409.786

```

Random-effects tobit regression

```

Number of obs      = 2,422
Uncensored         = 217
Left-censored      = 2,205
Right-censored     = 0

```

```

Limits: Lower = 0
Upper = +inf

```

```

Group variable: id
Random effects u_i ~ Gaussian

```

```

Number of groups   = 1,235
Obs per group:
    min = 1
    avg = 2.0
    max = 2

```

Integration method: **mvaghermite**

```

Integration pts.   = 12

```

Log likelihood = **-2409.786**

```

Wald chi2(9)       = 123.26
Prob > chi2        = 0.0000

```

```

> efficient
> Std. err.
> z
> P>|z|
> [95% con
> f. interval]

```

```

> 0
> (base)
> 1206.994
> 403.0685
> -2.99
> 0.003
> -1996.994
> -416.9942
>
> 0
> (base)
> 1072.297
> 770.8982
> 1.39
> 0.164
> -438.6353
> 2583.23
> Between 65 and 74 years old
> 2528.292
> 871.5916
> -2.90
> 0.004
> -4236.58
> -820.004
> Between 75 and 84 years old
> -3942.17
> 977.5494
> -4.03
> 0.000
> -5858.132
> -2026.208
> Older than 85 years
> 3311.749
> 1132.203
> -2.93
> 0.003
> -5530.827
> -1092.671
>
> 0
> (base)
> 3043.425
> 412.266
> 7.38
> 0.000
> 2235.398
> 3851.451
>
> Less than primary, primary and lower secondary education (ISCED levels 0-2)
> 0
> (base)

```

```

Upper secondary and post-secondary non-tertiary education (ISCED levels 3 and 4) |
> 388.2117
> 693.2673
> 0.56
> 0.575
> -970.5672
> 1746.991
> Tertiary education (ISCED levels 5-8) | -
> 138.1068
> 722.2035
> -0.19
> 0.848
> -1553.6
> 1277.386
> income | -
> .1073608
> .0270586
> -3.97
> 0.000
> -.1603946
> -.054327
> _cons | -
> 5432.212
> 1092.113
> -4.97
> 0.000
> -7572.715
> -3291.709
|-----|
> 3607.48 /sigma_u |
> 296.8856
> 12.15
> 0.000
> 3025.595
> 4189.365
> /sigma_e |
> 2226.034
> 167.9589
> 13.25
> 0.000
> 1896.841
> 2555.228
|-----|
> .7242368 rho |
> .0424687
> .6354625
> .8008308
|-----|

```

```

LR test of sigma_u=0: chibar2(01) = 117.37 Prob >= chibar2 = 0.000

```

```

135 outreg2 using results.xls, excel dec(3) alpha(0.01, 0.05, 0.10) symbol(**, **, *) c
> title(DisabilityLT) append
results.xls
dir : seeout

```

```

136 outreg2 using resultsNoStar.xls, excel dec(3) ctitle(DisabilityLT) noaster append
    resultsNoStar.xls
    dir : seeout
137
138 *for education
139
140 xttobit ypen3 i.gender i.ageE i.gali i.educ income if country==48, ll(0) iterate(25)
    > baselevels tobit

Fitting comparison model:

Fitting constant-only model:

Iteration 0: Log likelihood = -4852.3056
Iteration 1: Log likelihood = -2746.6214
Iteration 2: Log likelihood = -2638.7788
Iteration 3: Log likelihood = -2635.3474
Iteration 4: Log likelihood = -2635.3462
Iteration 5: Log likelihood = -2635.3462

Fitting full model:

Iteration 0: Log likelihood = -4765.2233
Iteration 1: Log likelihood = -2553.297
Iteration 2: Log likelihood = -2473.2867
Iteration 3: Log likelihood = -2470.0558
Iteration 4: Log likelihood = -2470.0473
Iteration 5: Log likelihood = -2470.0473

Obtaining starting values for full model:

Iteration 0: Log likelihood = -19437.74
Iteration 1: Log likelihood = -19429.848
Iteration 2: Log likelihood = -19429.823

Fitting full model:

Iteration 0: Log likelihood = -4422.2156
Iteration 1: Log likelihood = -2835.9785 (not concave)
Iteration 2: Log likelihood = -2478.9333 (not concave)
Iteration 3: Log likelihood = -2459.9196 (not concave)
Iteration 4: Log likelihood = -2449.4723 (not concave)
Iteration 5: Log likelihood = -2443.1141
Iteration 6: Log likelihood = -2414.0988
Iteration 7: Log likelihood = -2412.1468
Iteration 8: Log likelihood = -2412.1382
Iteration 9: Log likelihood = -2412.1373
Iteration 10: Log likelihood = -2412.1366
Iteration 11: Log likelihood = -2412.1366

Random-effects tobit regression                                Number of obs      = 2,422
                                                                Uncensored = 217
Limits: Lower = 0                                           Left-censored = 2,205
                                                                Right-censored = 0
                                                                Upper = +inf

Group variable: id                                           Number of groups   = 1,235
Random effects u_i ~ Gaussian                                Obs per group:
                                                                min = 1
                                                                avg = 2.0
                                                                max = 2

Integration method: mvaghermite                             Integration pts.    = 12

Log likelihood = -2412.1366                                Wald chi2(7)       = 121.75
                                                                Prob > chi2        = 0.0000

```

```

> efficient
> Std. err.
> z
> P>|z|
> [95% con
> f. interval]

> ypen3 | Co
> 0
> (base)
> 1240.203
> 399.6379
> -3.10
> 0.002
> -2023.479
> -456.9275
>
> ageE
> Less than 55 years old
> 0
> (base)
> 1067.015
> 775.6441
> 1.38
> 0.169
> -453.2194
> 2587.25
> Older than 64 years
> -3033.35
> 850.4555
> -3.57
> 0.000
> -4700.212
> -1366.488
>
> gali
> Not limited
> 0
> (base)
> 3052.411
> 413.8142
> 7.38
> 0.000
> 2241.35
> 3863.472
>
> educ
> Less than primary, primary and lower secondary education (ISCED levels 0-2)
> 0
> (base)
> Upper secondary and post-secondary non-tertiary education (ISCED levels 3 and 4)
> 797.0943
> 634.2378
> 1.26
> 0.209
> -445.9889
> 2040.177
> Tertiary education (ISCED levels 5-8)
> 234.42
> 670.1718
> 0.35
> 0.726
> -1079.093
> 1547.933

```

```

> .1088881                                income | -
>      .0272401
>      -4.00
>      0.000
>      -.1622776
>      -.0554985
>
> 5802.096                                _cons | -
>      1078.766
>      -5.38
>      0.000
>      -7916.438
>      -3687.754
+-----+-----+
> 3581.67                                /sigma_u |
>      292.1722
>      12.26
>      0.000
>      3009.023
>      4154.317
>
> 2259.991                                /sigma_e |
>      169.0658
>      13.37
>      0.000
>      1928.629
>      2591.354
+-----+-----+
> .7152329                                rho |
>      .042552
>      .626647
>      .7923176
+-----+-----+

```

LR test of sigma_u=0: chibar2(01) = 115.82 Prob >= chibar2 = 0.000

```

141 outreg2 using results.xls, excel dec(3) alpha(0.01, 0.05, 0.10) symbol(**, **, *) c
> title(EducationLT) append
  results.xls
  dir : seeout

```

```

142 outreg2 using resultsNoStar.xls, excel dec(3) ctitle(EducationLT) noaster append
  resultsNoStar.xls
  dir : seeout

```

```

143
144 *for income
145
146 xttobit ypen3 i.gender i.ageE i.gali i.educ income if country==48, ll(0) iterate(25)
> baselevels tobit

```

Fitting comparison model:

Fitting constant-only model:

```

Iteration 0: Log likelihood = -4852.3056
Iteration 1: Log likelihood = -2746.6214
Iteration 2: Log likelihood = -2638.7788
Iteration 3: Log likelihood = -2635.3474
Iteration 4: Log likelihood = -2635.3462
Iteration 5: Log likelihood = -2635.3462

```

Fitting full model:

Between 55 and 64 years old

```

> 1067.015
>      775.6441
>      1.38
>      0.169
>      -453.2194
>      2587.25
>      Older than 64 years |
> -3033.35
>      850.4555
>      -3.57
>      0.000
>      -4700.212
>      -1366.488
>
>      0
>      (base)
>      3052.411
>      413.8142
>      7.38
>      0.000
>      2241.35
>      3863.472
>
>      Less than primary, primary and lower secondary education (ISCED levels 0-2) educ |
>      0
>      (base)
> Upper secondary and post-secondary non-tertiary education (ISCED levels 3 and 4) |
> 797.0943
>      634.2378
>      1.26
>      0.209
>      -445.9889
>      2040.177
>      Tertiary education (ISCED levels 5-8) |
> 234.42
>      670.1718
>      0.35
>      0.726
>      -1079.093
>      1547.933
>
>      income | -
> .1088881
>      .0272401
>      -4.00
>      0.000
>      -.1622776
>      -.0554985
>
>      _cons | -
> 5802.096
>      1078.766
>      -5.38
>      0.000
>      -7916.438
>      -3687.754
>
>
>      /sigma_u |
> 3581.67
>      292.1722
>      12.26
>      0.000
>      3009.023
>      4154.317
>
>      /sigma_e |
> 2259.991
>      169.0658
>      13.37

```



```
>               0.000
>                1928.629
>                 2591.354
+-----+-----+-----+-----+
|                                     rho |
> .7152329
>      .042552
>              .626647
>             .7923176
+-----+-----+-----+-----+
LR test of sigma u=0: chibar2(01) = 115.82          Prob >= chibar2 = 0.000
```

```

Random-effects tobit regression
Limits: Lower = 0
        Upper = +inf
Group variable: id
Random effects u_i ~ Gaussian

Number of obs   = 2,003
Uncensored     = 132
Left-censored  = 1,871
Right-censored  = 0

Number of groups = 1,014
Obs per group:
    min = 1
    avg = 2.0
    max = 2

Integration method: mvaghermite
Integration pts. = 12

Log likelihood = -1431.668

Wald chi2(6) = 94.04
Prob > chi2 = 0.0000

```

						ypen3	Co
> efficient	Std. err.	z	P> z	[95% con	f. interval]		
						gender	
						Male	
> 0	(base)						
> 641.0127	402.2035	-1.59	0.111	-1429.317	147.2917	Female	-
						age	-
> 217.7174	29.76099	-7.32	0.000	-276.0479	-159.3869		
						gali	
						Not limited	
> 0	(base)					Limited	
> 4136.484	584.2676	7.08	0.000	2991.34	5281.627		
						educ	
						Less than primary, primary and lower secondary education (ISCED levels 0-2)	
> 0	(base)						
						Upper secondary and post-secondary non-tertiary education (ISCED levels 3 and 4)	
> 596.2011	616.5269	0.97	0.334	-612.1695	1804.572		
						Tertiary education (ISCED levels 5-8)	-
> 638.7354							

```

>      817.3516
>      -0.78
>      0.435
>      -2240.715
>      963.2443
>
>      income |
> -0.219487
>      .0454556
>      -4.83
>      0.000
>      -0.3085783
>      -0.1303956
>
>      _cons |
> 6995.528
>      1787.025
>      3.91
>      0.000
>      3493.023
>      10498.03
>
+-----+-----+
>                                     /sigma_u |
> 2633.705
>      288.8006
>      9.12
>      0.000
>      2067.666
>      3199.744
>
>                                     /sigma_e |
> 2104.673
>      191.4314
>      10.99
>      0.000
>      1729.475
>      2479.872
>
+-----+-----+
>                                     rho |
> .6102737
>      .0643687
>
>      .4805192
>      .7287097
>
+-----+-----+

```

LR test of sigma_u=0: chibar2(01) = 50.13 Prob >= chibar2 = 0.000

```

156 outreg2 using results.xls, excel dec(3) alpha(0.01, 0.05, 0.10) symbol(**, **, *) c
> title(Core) append
  results.xls
  dir : seeout

```

```

157 outreg2 using resultsNoStar.xls, excel dec(3) ctitle(Core) noaster append
  resultsNoStar.xls
  dir : seeout

```

```

158
159 *for disability
160

```

```
161 xtlogit yopen3 i.gender i.ageD i.gali i.educ income if country==57, ll(0) iterate(25)
> baselevels tobit
```

Fitting comparison model:

Fitting constant-only model:

```
Iteration 0: Log likelihood = -3579.9263
Iteration 1: Log likelihood = -1698.9624
Iteration 2: Log likelihood = -1616.2797
Iteration 3: Log likelihood = -1616.1126
Iteration 4: Log likelihood = -1616.1126
```

Fitting full model:

```
Iteration 0: Log likelihood = -3506.908
Iteration 1: Log likelihood = -1552.0409
Iteration 2: Log likelihood = -1457.4938
Iteration 3: Log likelihood = -1445.8596
Iteration 4: Log likelihood = -1445.3313
Iteration 5: Log likelihood = -1445.2677
Iteration 6: Log likelihood = -1445.2581
Iteration 7: Log likelihood = -1445.2563
Iteration 8: Log likelihood = -1445.2559
Iteration 9: Log likelihood = -1445.2559
Iteration 10: Log likelihood = -1445.2558
```

Obtaining starting values for full model:

```
Iteration 0: Log likelihood = -15445.47
Iteration 1: Log likelihood = -15441.736
Iteration 2: Log likelihood = -15441.727
```

Fitting full model:

```
Iteration 0: Log likelihood = -3290.4204
Iteration 1: Log likelihood = -1565.0715 (not concave)
Iteration 2: Log likelihood = -1487.9781
Iteration 3: Log likelihood = -1434.153
Iteration 4: Log likelihood = -1421.347
Iteration 5: Log likelihood = -1420.921
Iteration 6: Log likelihood = -1420.885
Iteration 7: Log likelihood = -1420.88
Iteration 8: Log likelihood = -1420.879
Iteration 9: Log likelihood = -1420.8788
Iteration 10: Log likelihood = -1420.8787
Iteration 11: Log likelihood = -1420.8787
```

Random-effects tobit regression

```
Limits: Lower = 0
        Upper = +inf
```

```
Group variable: id
Random effects u_i ~ Gaussian
```

Integration method: mvaghermite

Log likelihood = -1420.8787

```
Number of obs      = 2,003
      Uncensored    = 132
      Left-censored = 1,871
      Right-censored = 0
```

```
Number of groups   = 1,014
Obs per group:
      min = 1
      avg = 2.0
      max = 2
```

Integration pts. = 12

```
Wald chi2(9)      = 94.08
Prob > chi2       = 0.0000
```

```

> efficient
> Std. err.
> z
> P>|z|
> [95% con
> f. interval]

```

```

> 0
> (base)
> 341.5978
> 399.6168
> -0.85
> 0.393
> -1124.832
> 441.6368
>
> 0
> (base)
> 461.6804
> 593.136
> 0.78
> 0.436
> -700.8448
> 1624.206
> 2342.506
> 728.4104
> -3.22
> 0.001
> -3770.164
> -914.8481
> 5788.652
> 1271.983
> -4.55
> 0.000
> -8281.693
> -3295.611
> 15114.06
> 408420.4
> -0.04
> 0.970
> -815603.4
> 785375.3
>
> 0
> (base)
> 3705.149
> 543.4916
> 6.82
> 0.000
> 2639.925
> 4770.373
>
> 0
> (base)

```

```

ypen3 | Co
gender
Male |
Female | -
aged
Less than 55 years old |
Between 55 and 64 years old |
Between 65 and 74 years old | -
Between 75 and 84 years old | -
Older than 85 years | -
gali
Not limited |
Limited |
educ
Less than primary, primary and lower secondary education (ISCED levels 0-2)
0
(base)

```

```

Upper secondary and post-secondary non-tertiary education (ISCED levels 3 and 4) |
> 156.8724
> 617.0567
> 0.25
> 0.799
> -1052.536
> 1366.281
> Tertiary education (ISCED levels 5-8) | -
> 885.9491
> 815.3292
> -1.09
> 0.277
> -2483.965
> 712.0668
> income | -
> .1950166
> .0438524
> -4.45
> 0.000
> -.2809658
> -.1090674
> _cons | -
> 5421.555
> 990.1776
> -5.48
> 0.000
> -7362.267
> -3480.842
|-----|
| /sigma_u |
> 2570.351
> 279.8937
> 9.18
> 0.000
> 2021.769
> 3118.933
> /sigma_e |
> 2041.861
> 189.3496
> 10.78
> 0.000
> 1670.743
> 2412.98
|-----|
| rho |
> .6131
> .0653765
> .4811277
> .7330748
|-----|

```

LR test of sigma_u=0: chibar2(01) = 48.75

Prob >= chibar2 = 0.000

162 outreg2 using results.xls, excel dec(3) alpha(0.01, 0.05, 0.10) symbol(***, **, *) c
> title(DisabilityLV) append
results.xls
dir : seeout

```

163 outreg2 using resultsNoStar.xls, excel dec(3) ctitle(DisabilityLV) noaster append
    resultsNoStar.xls
    dir : seeout

```

```

164
165 *for education
166
167 xttobit ypen3 i.gender i.ageE i.gali i.educ income if country==57, ll(0) iterate(25)
    > baselevels tobit

```

Fitting comparison model:

Fitting constant-only model:

```

Iteration 0: Log likelihood = -3579.9263
Iteration 1: Log likelihood = -1698.9624
Iteration 2: Log likelihood = -1616.2797
Iteration 3: Log likelihood = -1616.1126
Iteration 4: Log likelihood = -1616.1126

```

Fitting full model:

```

Iteration 0: Log likelihood = -3511.125
Iteration 1: Log likelihood = -1555.4342
Iteration 2: Log likelihood = -1464.3483
Iteration 3: Log likelihood = -1455.0028
Iteration 4: Log likelihood = -1454.8874
Iteration 5: Log likelihood = -1454.8873

```

Obtaining starting values for full model:

```

Iteration 0: Log likelihood = -15450.207
Iteration 1: Log likelihood = -15446.684
Iteration 2: Log likelihood = -15446.676
Iteration 3: Log likelihood = -15446.676

```

Fitting full model:

```

Iteration 0: Log likelihood = -3293.0056
Iteration 1: Log likelihood = -1558.0001 (not concave)
Iteration 2: Log likelihood = -1483.8455
Iteration 3: Log likelihood = -1440.8292
Iteration 4: Log likelihood = -1431.4216
Iteration 5: Log likelihood = -1431.0339
Iteration 6: Log likelihood = -1431.0316
Iteration 7: Log likelihood = -1431.0316

```

Random-effects tobit regression

```

Limits: Lower = 0
        Upper = +inf

```

```

Group variable: id
Random effects u_i ~ Gaussian

```

Integration method: **mvaghermite**

Log likelihood = **-1431.0316**

```

Number of obs      = 2,003
Uncensored         = 132
Left-censored      = 1,871
Right-censored     = 0

```

```

Number of groups   = 1,014
Obs per group:
    min = 1
    avg = 2.0
    max = 2

```

Integration pts. = **12**

```

Wald chi2(7)       = 98.95
Prob > chi2        = 0.0000

```

```

> efficient
> Std. err.
> z
> P>|z|
> [95% con
> f. interval]

> ypen3 | Co
>
> 0
> (base)
> 433.5706
> 394.5748
> -1.10
> 0.272
> -1206.923
> 339.7818
>
> ageE
> Less than 55 years old
> 0
> (base)
> 383.5329
> 601.6318
> 0.64
> 0.524
> -795.6438
> 1562.71
> Older than 64 years
> -3431.92
> 739.1043
> -4.64
> 0.000
> -4880.537
> -1983.302
>
> gali
> Not limited
> 0
> (base)
> 3710.976
> 551.8659
> 6.72
> 0.000
> 2629.338
> 4792.613
>
> educ
> Less than primary, primary and lower secondary education (ISCED levels 0-2)
> 0
> (base)
> Upper secondary and post-secondary non-tertiary education (ISCED levels 3 and 4)
> 688.1303
> 582.771
> 1.18
> 0.238
> -454.0798
> 1830.34
> Tertiary education (ISCED levels 5-8)
> 468.8768
> 775.7078
> -0.60
> 0.546
> -1989.236
> 1051.483

```



```

> .1964301                                income | -
>      .0446491
>      -4.40
>      0.000
>      -.2839408
>      -.1089194
>
> 5825.039                                _cons | -
>      996.1594
>      -5.85
>      0.000
>      -7777.476
>      -3872.603
>
+-----+
> 2540.898                                /sigma_u |
>      275.857
>      9.21
>      0.000
>      2000.228
>      3081.568
>
> 2131.659                                /sigma_e |
>      194.8313
>      10.94
>      0.000
>      1749.796
>      2513.521
>
+-----+
> .5869169                                rho |
>      .064998
>      .4571951
>      .707723
>
+-----+
LR test of sigma_u=0: chibar2(01) = 47.71          Prob >= chibar2 = 0.000

```

```

168 outreg2 using results.xls, excel dec(3) alpha(0.01, 0.05, 0.10) symbol(**, **, *) c
> title(EducationLV) append
  results.xls
  dir : seeout

```

```

169 outreg2 using resultsNoStar.xls, excel dec(3) ctitle(EducationLV) noaster append
  resultsNoStar.xls
  dir : seeout

```

```

170
171 *for income
172
173 xttobit ypen3 i.gender i.ageE i.gali i.educ income if country==57, ll(0) iterate(25)
> baselevels tobit

```

Fitting comparison model:

Fitting constant-only model:

```

Iteration 0: Log likelihood = -3579.9263
Iteration 1: Log likelihood = -1698.9624
Iteration 2: Log likelihood = -1616.2797
Iteration 3: Log likelihood = -1616.1126
Iteration 4: Log likelihood = -1616.1126

```

Fitting full model:

```

> efficient
> Std. err.      z      P>|z|      [95% con
>                                     f. interval]
+-----+-----+-----+-----+-----+
>                                gender
>                                Male
> 0 (base)
>                                Female
> 433.5706    394.5748    -1.10    0.272    -1206.923    339.7818
>
>                                ageE
>                                Less than 55 years old
> 0 (base)
>                                Between 55 and 64 years old
> 383.5329    601.6318    0.64
>

```



```

> .5869169
>
> .064998
>
> .4571951
>
> .707723
>

```

```

LR test of sigma_u=0: chibar2(01) = 47.71 Prob >= chibar2 = 0.000

```

```

74 outreg2 using results.xls, excel dec(3) alpha(0.01, 0.05, 0.10) symbol(***, **, *) c
> title(IncomeLV) append
results.xls
dir : seeout

```

```

75 outreg2 using resultsNoStar.xls, excel dec(3) ctitle(IncomeLV) noaster append
resultsNoStar.xls
dir : seeout

```

```

76
77 log close SimulationBaltic
   name: SimulationBaltic
   log:  C:\Users\hansg\Documents\bestanden\PhD ebs\JRF work Stata\SHARE data 7\ou
> tput.smcl
   log type: smcl
   closed on: 25 Aug 2026, 11:20:50

```